

Course Map

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1. Overview

Key point: The Course map ties every Note together. It shows where each idea sits in an ML project, how ideas connect, what to read first, and which algorithm to choose.

The Notes each teach one lesson. The **Course map** (G-2160) shows how those lessons fit together, in four views:

View	Question it answers
Pipeline map (G-2161)	Where does this idea sit in a real ML project?
Concept map (G-2162)	How is this idea connected to the others?
Learning path (G-2163)	Which Notes should I read first?
Algorithm chooser (G-2164)	Which algorithm suits my problem?

Each idea on the map is a **Concept** (G-2165). Concepts already taught in a written Note are **confirmed**; the others are **draft** (G-2167), shown faint, and are checked against their source when their Note is written. So far, 335 of 335 Concepts are confirmed.

Every Note starts with a *Where this fits* box: a small Pipeline map with that Note's steps highlighted, plus what it builds on and what it leads to.

Extra: This map is generated from one data file, `course_map/concepts.yaml`. An interactive version, where you can zoom, filter by step and click a Concept to see its links and Notes, runs with `python course_map/app.py`.

2. The Pipeline map

Key point: Every ML project moves through the same steps, from framing the problem to monitoring the deployed model. Every Concept belongs to one step.

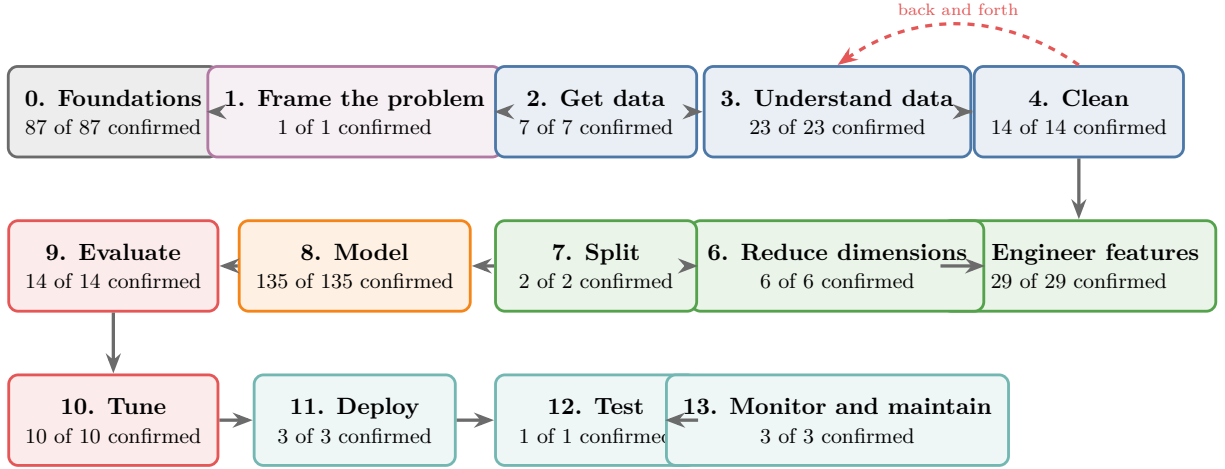


Figure 1: The Pipeline map: 14 steps of an ML project

Figure 1 shows the 14 steps. They follow the ML development life cycle, or **MLDLC** (G-1240):

1. frame the problem;
2. get the data and understand it;
3. clean the data;
4. engineer features;
5. train and evaluate models;
6. tune them;
7. deploy, test and keep the model healthy.

Steps 3 (Understand data) and 4 (Clean) loop: exploring reveals what needs cleaning, and cleaning changes the data, so we explore again.

2.1 Step 0: Foundations

Concept	Taught in	Status
Machine learning	Note ML-001 , Note ML-002	confirmed
Data mining	Note ML-001 , Note ML-008	confirmed
Artificial intelligence	Note ML-002	confirmed
Symbolic AI and expert systems	Note ML-002	confirmed
Deep learning	Note ML-002	confirmed
Neural networks	Note ML-002 , Note DL-002	confirmed
Features	Note ML-002 , Note ML-010	confirmed
Supervised learning	Note ML-003	confirmed
Regression problems	Note ML-003	confirmed
Classification problems	Note ML-003	confirmed

Concept	Taught in	Status
Unsupervised learning	Note ML-003	confirmed
Semi-supervised learning	Note ML-003	confirmed
Reinforcement learning	Note ML-003	confirmed
Batch (offline) learning	Note ML-004	confirmed
Online learning	Note ML-005	confirmed
Out-of-core learning	Note ML-005	confirmed
Instance-based learning	Note ML-006	confirmed
Model-based learning	Note ML-006	confirmed
Applications of ML	Note ML-008	confirmed
ML development life cycle	Note ML-009 , Note ML-012	confirmed
Tensors	Note ML-010	confirmed
Setup: conda, Jupyter and Colab	Note ML-011	confirmed
Chi-square tests	Note MA-045 , Note ML-028	confirmed
Vector magnitude, distance and scalar operations	Note ML-038 , Note MA-049	confirmed
Normal distribution	Note ML-041 , Note ML-084 , Note MA-020 , Note MA-024 , Note MA-069	confirmed
Eigenvectors and eigenvalues	Note ML-047 , Note MA-056	confirmed
Dot product	Note ML-047 , Note MA-050 , Note MA-055	confirmed
Linear transformations and matrices	Note ML-047 , Note MA-053	confirmed
Derivatives of one variable	Note ML-050 , Note MA-061	confirmed
Equation of a hyperplane	Note ML-052 , Note ML-069 , Note MA-051	confirmed
Partial derivatives and gradients	Note ML-056 , Note MA-062	confirmed
Conditional probability	Note MA-015 , Note MA-014	confirmed
Independent and mutually exclusive events	Note MA-016 , Note MA-017	confirmed
Bayes' theorem	Note MA-018 , Note MA-019 , Note MA-014	confirmed
Bernoulli and binomial distributions	Note ML-096 , Note MA-021 , Note MA-031	confirmed
Hessian and multivariate Taylor	Note ML-120 , Note MA-064	confirmed
Taylor series	Note ML-120 , Note MA-061 , Note MA-064	confirmed
Probability distributions	Note MA-003 , Note MA-020	confirmed
Inferential statistics	Note MA-003 , Note MA-004	confirmed

Concept	Taught in	Status
Population, sample, parameter and statistic	Note MA-004	confirmed
Random variables	Note MA-020	confirmed
Probability mass function (PMF)	Note MA-021	confirmed
Uniform distribution	Note MA-021 , Note MA-029	confirmed
Log-normal distribution	Note MA-022 , Note MA-029	confirmed
Poisson distribution	Note MA-022 , Note MA-032	confirmed
Standard normal and the z-table	Note MA-025	confirmed
Pareto distribution and power laws	Note MA-030	confirmed
Sampling distribution and standard error	Note MA-033	confirmed
Central limit theorem	Note MA-033 , Note MA-034	confirmed
Confidence intervals	Note MA-035 , Note MA-036 , Note MA-037	confirmed
Student's t-distribution	Note MA-037	confirmed
Hypothesis testing: null and alternative	Note MA-038 , Note MA-039	confirmed
Z-test and rejection regions	Note MA-039	confirmed
Type I and II errors, power, tails	Note MA-040	confirmed
P-values	Note MA-041	confirmed
T-tests: one-sample, two-sample, paired	Note MA-042 , Note MA-043 , Note MA-044	confirmed
Events and sample spaces	Note MA-010	confirmed
Empirical vs theoretical probability, probability rules	Note MA-011 , Note MA-013	confirmed
Expected value and variance of a random variable	Note MA-012	confirmed
Venn diagrams and contingency tables	Note MA-013	confirmed
Joint and marginal probability	Note MA-014	confirmed
Linear algebra roadmap	Note MA-047	confirmed
Vectors and feature vectors	Note MA-048	confirmed
Role of mathematics in ML	Note MA-001	confirmed
Linear combinations, span and basis	Note MA-052	confirmed
Matrix multiplication as composition	Note MA-054	confirmed
Determinant	Note MA-056	confirmed

Concept	Taught in	Status
Choosing a hypothesis test	Note MA-044	confirmed
One-sample proportion test	Note MA-044	confirmed
One-way ANOVA	Note MA-046	confirmed
How to learn the maths for ML	Note MA-002	confirmed
Jacobian and matrix gradients	Note MA-063	confirmed
Singular value decomposition	Note MA-057 , Note MA-058	confirmed
Lagrange multipliers, KKT and duality	Note MA-066	confirmed
Convex sets and convex optimisation	Note MA-067	confirmed
Linear and quadratic programming	Note MA-068	confirmed
Likelihood	Note MA-069	confirmed
Maximum likelihood estimation (MLE)	Note MA-070 , Note MA-071 , Note MA-072	confirmed
Exponential distribution	Note MA-071	confirmed
Multivariate normal distribution	Note MA-073	confirmed
What deep learning is	Note DL-001 , Note DL-002	confirmed
Representation learning	Note DL-002	confirmed
History of deep learning	Note DL-003	confirmed
Universal approximation theorem	Note DL-003 , Note DL-009	confirmed
Memoization	Note DL-019	confirmed
Exponentially weighted moving average (EWMA)	Note DL-033	confirmed
Sequential data	Note DL-055	confirmed

2.2 Step 1: Frame the problem

Concept	Taught in	Status
Framing an ML problem	Note ML-009 , Note ML-013	confirmed

2.3 Step 2: Get data

Concept	Taught in	Status
Labelled data	Note ML-003 , Note ML-007	confirmed
Enough data	Note ML-007	confirmed

Concept	Taught in	Status
Sampling noise and bias	Note ML-007	confirmed
APIs	Note ML-007 , Note ML-009 , Note ML-016	confirmed
Web scraping	Note ML-007 , Note ML-009 , Note ML-017	confirmed
CSV files	Note ML-009 , Note ML-012 , Note ML-014	confirmed
JSON and SQL data	Note ML-015	confirmed

2.4 Step 3: Understand data

Concept	Taught in	Status
Exploratory data analysis	Note ML-009 , Note ML-012 , Note ML-018	confirmed
Univariate analysis	Note ML-009 , Note ML-019	confirmed
Bivariate and multivariate analysis	Note ML-009 , Note ML-020	confirmed
Imbalanced data	Note ML-009 , Note ML-127	confirmed
Variance	Note ML-018 , Note ML-046 , Note MA-006	confirmed
Correlation	Note ML-018 , Note ML-020 , Note MA-009	confirmed
Descriptive statistics	Note ML-018 , Note MA-003 , Note MA-005 , Note MA-006 , Note MA-007 , Note MA-008	confirmed
Probability density function (PDF)	Note ML-019 , Note ML-084 , Note MA-022	confirmed
Kernel density estimation (KDE)	Note ML-019 , Note MA-023 , Note MA-027	confirmed
Skewness	Note ML-019 , Note MA-026	confirmed
Kurtosis and moments	Note ML-021 , Note MA-028	confirmed
Pandas Profiling	Note ML-021	confirmed
Q-Q plot	Note ML-029 , Note MA-028	confirmed
Covariance and covariance matrix	Note ML-047 , Note MA-009	confirmed
Discrete and continuous data	Note MA-004	confirmed
Measures of central tendency	Note MA-005	confirmed
Bessel's correction	Note MA-006	confirmed
Frequency tables	Note MA-007	confirmed
Percentiles, quartiles and box plots	Note MA-008	confirmed
Correlation and causation	Note MA-009	confirmed

Concept	Taught in	Status
Cumulative distribution function (CDF)	Note MA-021 , Note MA-022 , Note MA-027	confirmed
Density estimation	Note MA-023	confirmed
Correlation significance test	Note MA-044	confirmed

2.5 Step 4: Clean

Concept	Taught in	Status
Poor-quality data	Note ML-007 , Note ML-009	confirmed
Missing values	Note ML-007 , Note ML-009 , Note ML-022 , Note ML-034 , Note ML-035 , Note ML-036 , Note ML-037 , Note ML-117	confirmed
Outliers	Note ML-007 , Note ML-009 , Note ML-019 , Note ML-022 , Note ML-040	confirmed
Simple imputation (mean, median, mode, constant)	Note ML-022 , Note ML-035 , Note ML-036	confirmed
Complete case analysis	Note ML-034	confirmed
Missing indicator	Note ML-037	confirmed
Random sample imputation	Note ML-037	confirmed
KNN imputer	Note ML-038	confirmed
Iterative imputation (MICE)	Note ML-039	confirmed
Trimming outliers	Note ML-040 , Note ML-041 , Note ML-042 , Note ML-043	confirmed
Capping (winsorization)	Note ML-040 , Note ML-041 , Note ML-042 , Note ML-043	confirmed
Z-score outlier method	Note ML-040 , Note ML-041 , Note MA-025	confirmed
IQR outlier method	Note ML-040 , Note ML-042	confirmed
Percentile outlier method	Note ML-040 , Note ML-043	confirmed

2.6 Step 5: Engineer features

Concept	Taught in	Status
Feature construction and splitting	Note ML-003 , Note ML-007 , Note ML-009 , Note ML-022 , Note ML-044	confirmed
Feature scaling	Note ML-006 , Note ML-009 , Note ML-012 , Note ML-022 , Note ML-023 , Note DL-023	confirmed

Concept	Taught in	Status
Feature engineering	Note ML-007 , Note ML-009 , Note ML-022	confirmed
Feature selection	Note ML-009 , Note ML-012 , Note ML-022 , Note ML-045	confirmed
Standardization	Note ML-009 , Note ML-012 , Note ML-023 , Note DL-023 , Note DL-031	confirmed
One-hot encoding	Note ML-010 , Note ML-022 , Note ML-026	confirmed
ML pipelines	Note ML-012 , Note ML-028 , Note ML-037	confirmed
Encoding categorical data	Note ML-022 , Note ML-025	confirmed
Binning and binarization	Note ML-022 , Note ML-031	confirmed
Feature transformation	Note ML-022	confirmed
Normalization	Note ML-024	confirmed
Ordinal and label encoding	Note ML-025	confirmed
Column transformer	Note ML-027	confirmed
Function transformer	Note ML-029	confirmed
Power transformer	Note ML-030	confirmed
Mixed variables	Note ML-032	confirmed
Date and time features	Note ML-033	confirmed
Feature importance	Note ML-093 , Note ML-094 , Note ML-108	confirmed
Permutation importance	Note ML-108	confirmed
Random under- and oversampling	Note ML-127	confirmed
SMOTE	Note ML-127	confirmed
Bag of words	Note MA-048	confirmed
Scaling inputs for neural networks	Note DL-023	confirmed
Data augmentation	Note DL-050	confirmed
Sequence padding	Note DL-055 , Note DL-057	confirmed
Tokenization and integer encoding of text	Note DL-057	confirmed
Word embeddings	Note DL-057	confirmed
Contextual embeddings	Note DL-073	confirmed
Meaning as direction in embedding space	Note DL-072	confirmed

2.7 Step 6: Reduce dimensions

Concept	Taught in	Status
Dimensionality reduction	Note ML-003 , Note ML-045	confirmed
PCA	Note ML-003 , Note ML-046 , Note ML-047 , Note ML-048	confirmed
Feature extraction	Note ML-022 , Note ML-045 , Note ML-046	confirmed
Curse of dimensionality	Note ML-045 , Note ML-085	confirmed
Low-rank approximation (truncated SVD)	Note MA-059	confirmed
Latent semantic analysis	Note MA-060	confirmed

2.8 Step 7: Split

Concept	Taught in	Status
Train-test split	Note ML-012	confirmed
Data leakage	Note ML-012 , Note ML-085	confirmed

2.9 Step 8: Model

Concept	Taught in	Status
Clustering	Note ML-003 , Note ML-122 , Note ML-123 , Note ML-124 , Note ML-125 , Note ML-126	confirmed
Anomaly detection	Note ML-003 , Note ML-126	confirmed
Association rule learning	Note ML-003	confirmed
Stochastic gradient descent	Note ML-005 , Note ML-058 , Note DL-020	confirmed
K-nearest neighbours	Note ML-006 , Note ML-085	confirmed
Ensemble learning	Note ML-009 , Note ML-095	confirmed
Logistic regression	Note ML-012 , Note ML-069 , Note ML-070 , Note ML-071 , Note ML-072 , Note ML-074	confirmed
Multicollinearity	Note ML-026	confirmed
K-means	Note ML-031 , Note ML-122 , Note ML-123 , Note ML-124 , Note MA-074	confirmed
Simple linear regression	Note ML-049 , Note ML-050	confirmed
Best-fit line and squared error	Note ML-049 , Note ML-050	confirmed
Ordinary least squares (closed form)	Note ML-050	confirmed

Concept	Taught in	Status
Multiple linear regression	Note ML-052 , Note ML-053 , Note ML-054	confirmed
Normal equation	Note ML-053 , Note ML-054	confirmed
Assumptions of linear regression	Note ML-055	confirmed
Gradient descent	Note ML-056 , Note DL-017 , Note DL-020	confirmed
Convex and non-convex loss	Note ML-056 , Note MA-065 , Note DL-017	confirmed
Batch gradient descent	Note ML-057 , Note DL-020	confirmed
Mini-batch gradient descent	Note ML-059 , Note DL-020	confirmed
Polynomial regression	Note ML-060	confirmed
Polynomial features	Note ML-060 , Note ML-079	confirmed
Regularisation	Note ML-062 , Note DL-026	confirmed
Ridge regression	Note ML-062 , Note ML-063 , Note ML-064 , Note ML-065	confirmed
Lasso regression	Note ML-066 , Note ML-067	confirmed
Elastic Net	Note ML-068	confirmed
Perceptron trick	Note ML-069 , Note ML-070 , Note DL-005	confirmed
Sigmoid function	Note ML-071 , Note ML-073 , Note DL-018 , Note DL-027	confirmed
Log loss (binary cross entropy)	Note ML-072 , Note DL-014 , Note MA-072	confirmed
Softmax regression	Note ML-078	confirmed
Naive Bayes	Note ML-081 , Note ML-082 , Note ML-083 , Note ML-084	confirmed
Support vector machines	Note ML-086 , Note ML-087 , Note ML-088	confirmed
Hinge loss and soft margin	Note ML-088	confirmed
Kernel trick	Note ML-089 , Note ML-090	confirmed
Entropy, information gain and Gini	Note ML-091	confirmed
Decision trees	Note ML-091 , Note ML-092 , Note ML-094	confirmed
Regression trees	Note ML-093	confirmed
Boosting	Note ML-095 , Note ML-113	confirmed
Voting ensembles	Note ML-096 , Note ML-097 , Note ML-098	confirmed
Bagging	Note ML-099 , Note ML-100 , Note ML-101	confirmed

Concept	Taught in	Status
Random forest	Note ML-102 , Note ML-103 , Note ML-104 , Note ML-105 , Note ML-106 , Note ML-107 , Note ML-108	confirmed
AdaBoost	Note ML-109 , Note ML-110 , Note ML-111 , Note ML-112	confirmed
Gradient boosting	Note ML-114 , Note ML-115 , Note ML-116	confirmed
XGBoost	Note ML-117 , Note ML-118 , Note ML-119 , Note ML-120	confirmed
Stacking and blending	Note ML-121	confirmed
Hierarchical clustering	Note ML-125	confirmed
DBSCAN	Note ML-126	confirmed
Balanced random forest	Note ML-127	confirmed
Cost-sensitive learning	Note ML-127	confirmed
Cosine similarity	Note MA-050	confirmed
Moore-Penrose pseudo-inverse	Note MA-060	confirmed
Categorical and sparse categorical cross-entropy	Note DL-012 , Note DL-014 , Note MA-072	confirmed
MAP estimation	Note MA-072	confirmed
Gaussian mixture model (GMM)	Note MA-073 , Note MA-074	confirmed
Expectation maximization (EM)	Note MA-074	confirmed
Types of neural networks	Note DL-003	confirmed
Multi-layer perceptron (MLP)	Note DL-003 , Note DL-009	confirmed
Perceptron	Note DL-004	confirmed
Perceptron loss	Note DL-006	confirmed
Problem with the perceptron (XOR)	Note DL-007	confirmed
MLP notation and parameter count	Note DL-008	confirmed
Forward propagation	Note DL-010	confirmed
Keras workflow	Note DL-011 , Note DL-012 , Note DL-013 , Note DL-022 , Note DL-025 , Note DL-026	confirmed
ANN for classification	Note DL-011 , Note DL-012	confirmed
ANN for regression	Note DL-013	confirmed
Loss functions in deep learning	Note DL-014	confirmed
Huber loss	Note DL-014	confirmed

Concept	Taught in	Status
Backpropagation	Note DL-015 , Note DL-016 , Note DL-017 , Note DL-019	confirmed
Vanishing gradient	Note DL-018 , Note DL-029 , Note DL-030 , Note DL-060	confirmed
Exploding gradient and gradient clipping	Note DL-018 , Note DL-029 , Note DL-060	confirmed
ReLU	Note DL-018 , Note DL-027 , Note DL-028	confirmed
Batch size in Keras	Note DL-020	confirmed
Dropout	Note DL-024 , Note DL-025	confirmed
L1 and L2 regularisation in neural networks	Note DL-026	confirmed
Activation functions	Note DL-027	confirmed
Tanh	Note DL-027	confirmed
Dying ReLU problem	Note DL-028	confirmed
Leaky ReLU, PReLU, ELU and SELU	Note DL-028	confirmed
Weight initialisation	Note DL-029 , Note DL-030	confirmed
Xavier and He initialisation	Note DL-030	confirmed
Batch normalisation	Note DL-031	confirmed
Covariate shift	Note DL-031	confirmed
Optimizers in deep learning	Note DL-032	confirmed
Local minima and saddle points	Note DL-032 , Note DL-034	confirmed
SGD with momentum	Note DL-034	confirmed
Nesterov accelerated gradient (NAG)	Note DL-035	confirmed
AdaGrad	Note DL-036	confirmed
RMSProp	Note DL-037	confirmed
Adam	Note DL-038	confirmed
Convolutional neural network (CNN)	Note DL-040 , Note DL-041 , Note DL-046	confirmed
Convolution operation and feature maps	Note DL-042	confirmed
Padding and strides	Note DL-043	confirmed
Pooling	Note DL-044	confirmed
CNN architecture (LeNet-5)	Note DL-045	confirmed
Backpropagation in a CNN	Note DL-047 , Note DL-048	confirmed
Image classification with a CNN (cats vs dogs)	Note DL-049	confirmed
Pretrained models and ImageNet	Note DL-051	confirmed

Concept	Taught in	Status
Transfer learning (feature extraction and fine-tuning)	Note DL-053	confirmed
Keras functional API	Note DL-054	confirmed
Skip connections	Note DL-054	confirmed
Recurrent neural network (RNN)	Note DL-055 , Note DL-056 , Note DL-057	confirmed
Parameter sharing across time steps	Note DL-056 , Note DL-059	confirmed
Types of RNN (many-to-one, one-to-many, many-to-many)	Note DL-058	confirmed
Sequence-to-sequence (encoder-decoder)	Note DL-058 , Note DL-068	confirmed
Backpropagation through time (BPTT)	Note DL-059	confirmed
Long-term dependency problem	Note DL-060	confirmed
LSTM (long short-term memory)	Note DL-061 , Note DL-062 , Note DL-063	confirmed
LSTM gates (forget, input, output) and cell state	Note DL-062	confirmed
Next-word prediction with an LSTM	Note DL-063	confirmed
GRU (gated recurrent unit)	Note DL-064	confirmed
Deep (stacked) RNNs	Note DL-065	confirmed
Bidirectional RNNs	Note DL-066	confirmed
Large language models (LLMs)	Note DL-067	confirmed
Teacher forcing	Note DL-068 , Note DL-082	confirmed
Attention mechanism	Note DL-069	confirmed
Bahdanau (additive) attention	Note DL-069 , Note DL-070	confirmed
Luong (multiplicative) attention	Note DL-070	confirmed
Transformer	Note DL-071 , Note DL-081 , Note DL-084	confirmed
Self-attention (query, key, value)	Note DL-073 , Note DL-074 , Note DL-076 , Note DL-077	confirmed
Scaled dot-product attention	Note DL-075	confirmed
Multi-head attention	Note DL-078	confirmed
Positional encoding	Note DL-079	confirmed
Layer normalisation	Note DL-080	confirmed
Residual connections and add & norm	Note DL-081	confirmed

Concept	Taught in	Status
Transformer encoder	Note DL-081	confirmed
Masked self-attention	Note DL-082	confirmed
Cross-attention	Note DL-083	confirmed
Transformer decoder	Note DL-084	confirmed
Transformer inference (autoregressive decoding, KV cache, beam search)	Note DL-085	confirmed
The transformer end to end (capstone)	Note DL-086	confirmed
Label smoothing	Note DL-086	confirmed
Decoder-only GPT	Note DL-087	confirmed
GELU activation	Note DL-087	confirmed
Unembedding, logits, temperature and sampling	Note DL-088	confirmed
MLP blocks as fact storage	Note DL-089	confirmed
Superposition and nearly perpendicular directions	Note DL-090	confirmed

2.10 Step 9: Evaluate

Concept	Taught in	Status
Overfitting	Note ML-007 , Note ML-060 , Note ML-085 , Note DL-026	confirmed
Underfitting	Note ML-007 , Note ML-060 , Note ML-085	confirmed
Accuracy	Note ML-012 , Note ML-075 , Note ML-085	confirmed
Cross-validation	Note ML-028 , Note ML-029 , Note ML-085 , Note ML-097 , Note ML-098 , Note ML-106 , Note ML-112 , Note ML-121	confirmed
Regression metrics	Note ML-051	confirmed
Bias-variance trade-off	Note ML-061 , Note ML-103	confirmed
Confusion matrix	Note ML-075	confirmed
Precision, recall and F1	Note ML-076	confirmed
ROC curve and AUC	Note ML-077	confirmed
Decision surface and boundary	Note ML-085	confirmed
OOB score	Note ML-099 , Note ML-100 , Note ML-101 , Note ML-107	confirmed
Training curves (History)	Note DL-011 , Note DL-012 , Note DL-013 , Note DL-022	confirmed

Concept	Taught in	Status
Visualising what a CNN learns	Note DL-052	confirmed
Logit lens	Note DL-088	confirmed

2.11 Step 10: Tune

Concept	Taught in	Status
Hyperparameter tuning	Note ML-009 , Note ML-080 , Note ML-085 , Note ML-092 , Note ML-105 , Note ML-112 , Note DL-021 , Note DL-039	confirmed
Grid and random search	Note ML-028 , Note ML-037 , Note ML-085 , Note ML-093 , Note ML-100 , Note ML-101 , Note ML-106 , Note ML-112	confirmed
Learning rate	Note ML-056 , Note ML-112 , Note ML-114 , Note DL-017	confirmed
Elbow method and WCSS	Note ML-122 , Note ML-123	confirmed
Bayesian optimisation	Note ML-128	confirmed
Optuna	Note ML-128	confirmed
Improving a neural network	Note DL-021	confirmed
Early stopping	Note DL-021 , Note DL-022	confirmed
Keras Tuner	Note DL-039	confirmed
Learning-rate warm-up schedule	Note DL-086	confirmed

2.12 Step 11: Deploy

Concept	Taught in	Status
Deployment	Note ML-004 , Note ML-007 , Note ML-009 , Note ML-012 , Note ML-028	confirmed
Software integration	Note ML-007	confirmed
Saving models with pickle	Note ML-009 , Note ML-012 , Note ML-028	confirmed

2.13 Step 12: Test

Concept	Taught in	Status
Beta and A/B testing	Note ML-009 , Note MA-040	confirmed

2.14 Step 13: Monitor and maintain

Concept	Taught in	Status
Model drift	Note ML-004 , Note ML-009	confirmed
Retraining	Note ML-004 , Note ML-005 , Note ML-009	confirmed
MLOps and cost	Note ML-007 , Note ML-009	confirmed

3. The Concept map

Key point: Concepts are joined by five kinds of Link. Following the Links shows why each idea exists and what it is for.

Every **Link** (G-2166) has one of five types:

Link	Meaning	Example
needs	must be understood first	logistic regression <i>needs</i> gradient descent
is a kind of	a special case	Ridge <i>is a kind of</i> regularisation
fixes	solves a problem	regularisation <i>fixes</i> overfitting
compared with	often confused or contrasted	bagging <i>compared with</i> boosting
used in	a tool used inside something else	feature scaling <i>used in</i> KNN

The full map has 335 Concepts, too many for one page, so each topic has its own mind map (Figures 2 to 19). Each map tells the topic's story from left to right and shows only its key Concepts and links; the grey number in each box is the Note that teaches it, and dashed boxes lead to other maps. To see every Concept and every link at once, open `course_map/concept_map_3d.html` in a browser: a 3D network you can rotate and zoom, with one colour per area (rebuilt by `python course_map/map_3d.py`). The interactive app (`python course_map/app.py`) lists each Concept's links with the Notes that teach them.

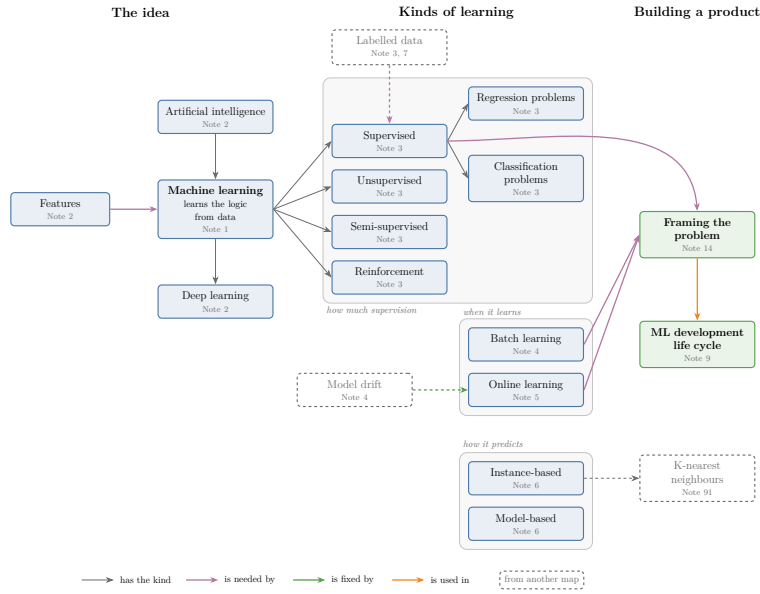


Figure 2: Concept map: Foundations and framing

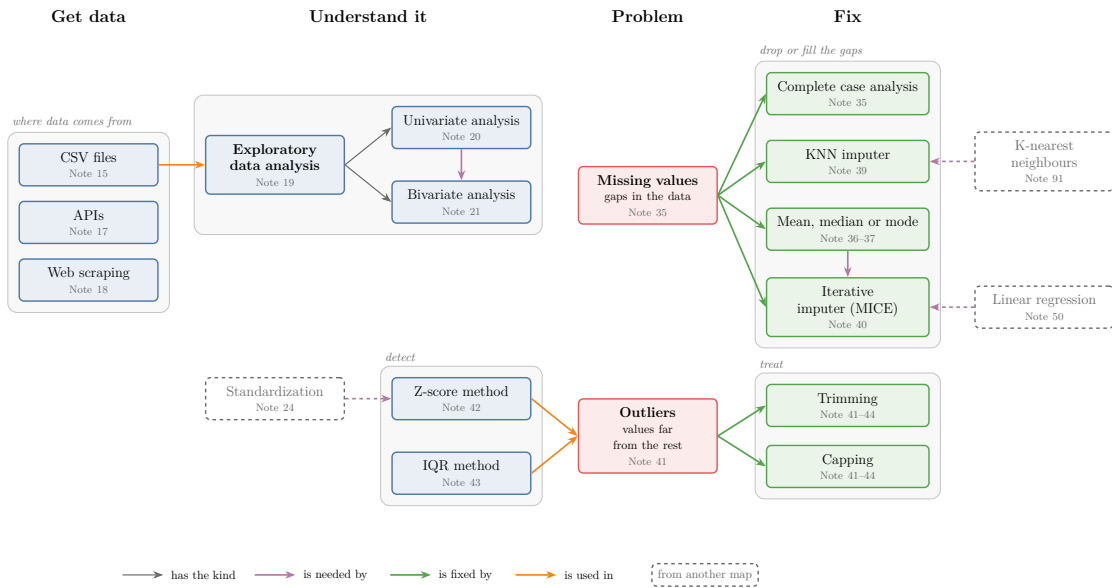


Figure 3: Concept map: Getting, understanding and cleaning data

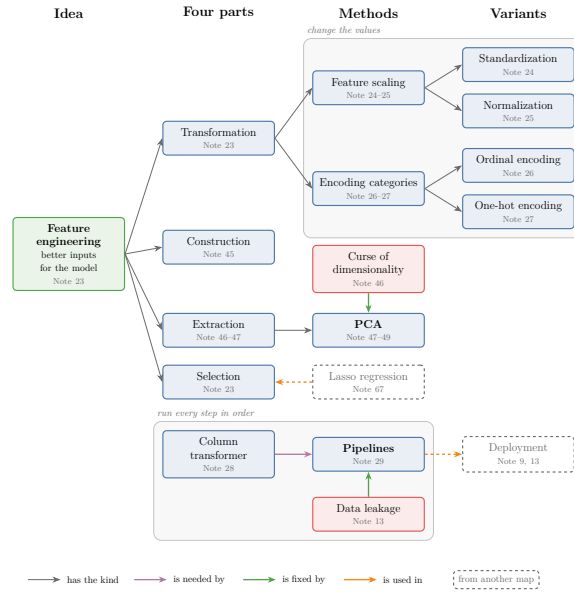


Figure 4: Concept map: Features, dimensions and splitting

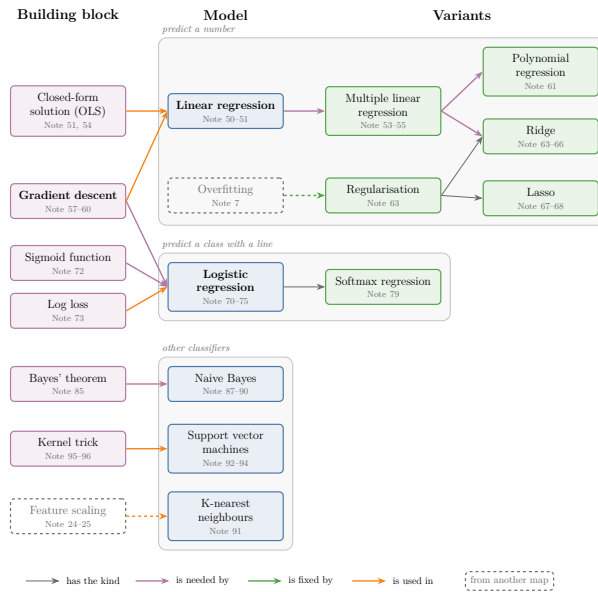


Figure 5: Concept map: Models: regression, classification and gradient descent

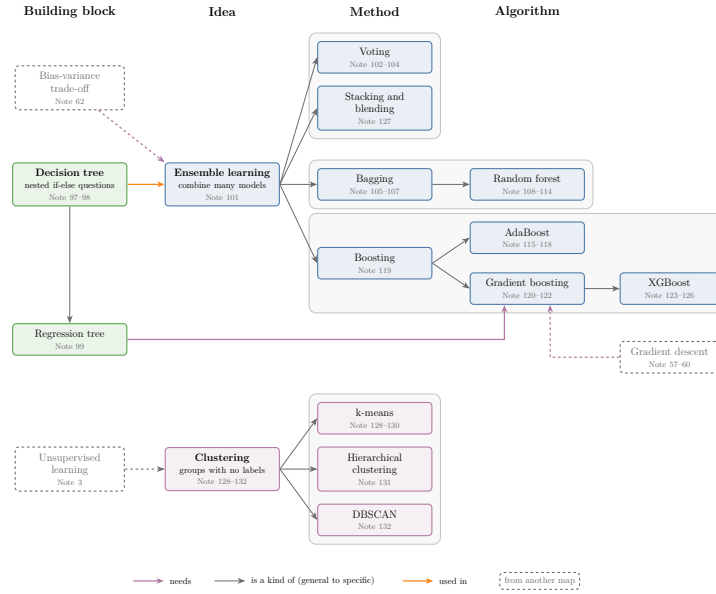


Figure 6: Concept map: Models: trees, ensembles and clustering

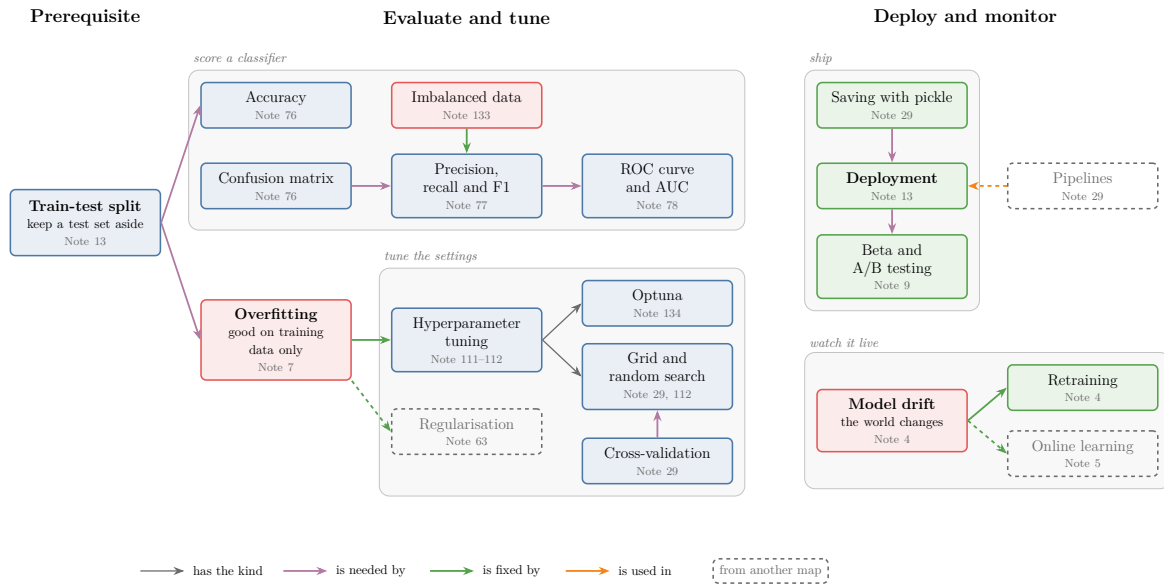


Figure 7: Concept map: Evaluating, tuning and production

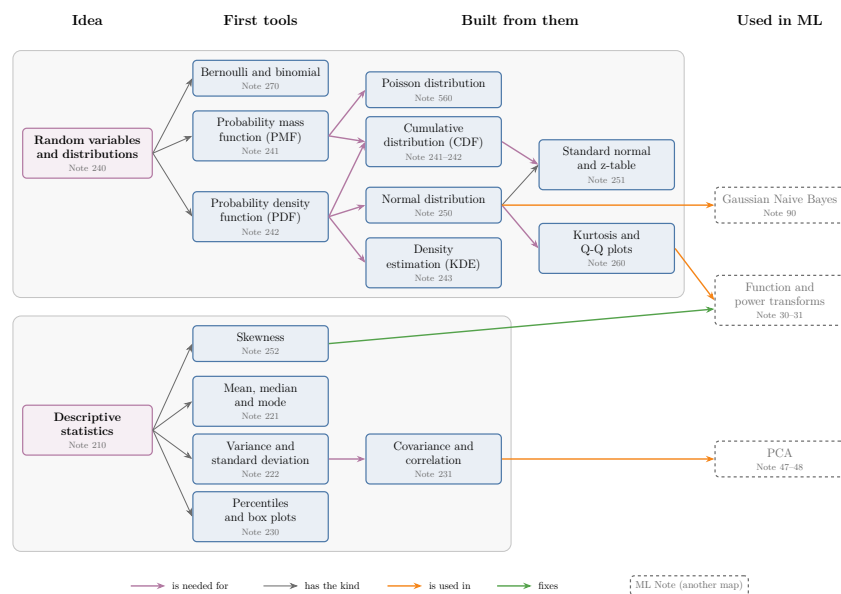


Figure 8: Concept map: Descriptive statistics and distributions

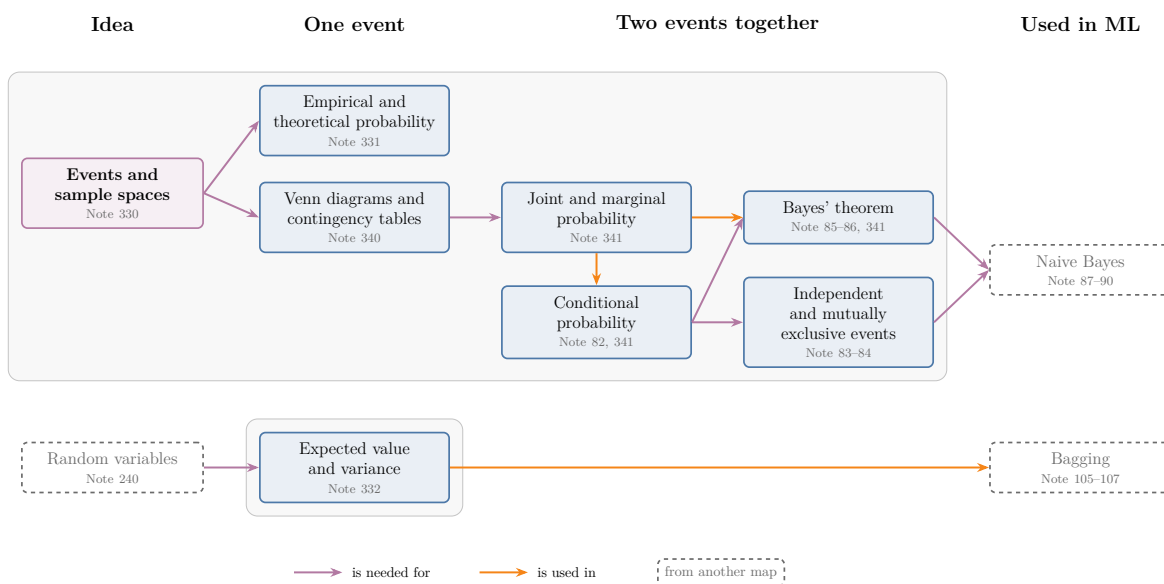


Figure 9: Concept map: Probability

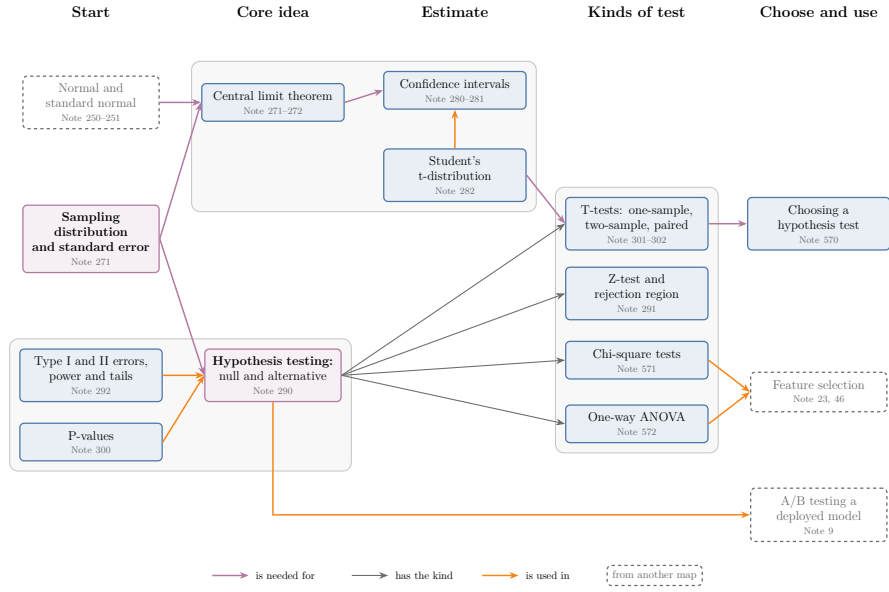


Figure 10: Concept map: Sampling, inference and hypothesis tests

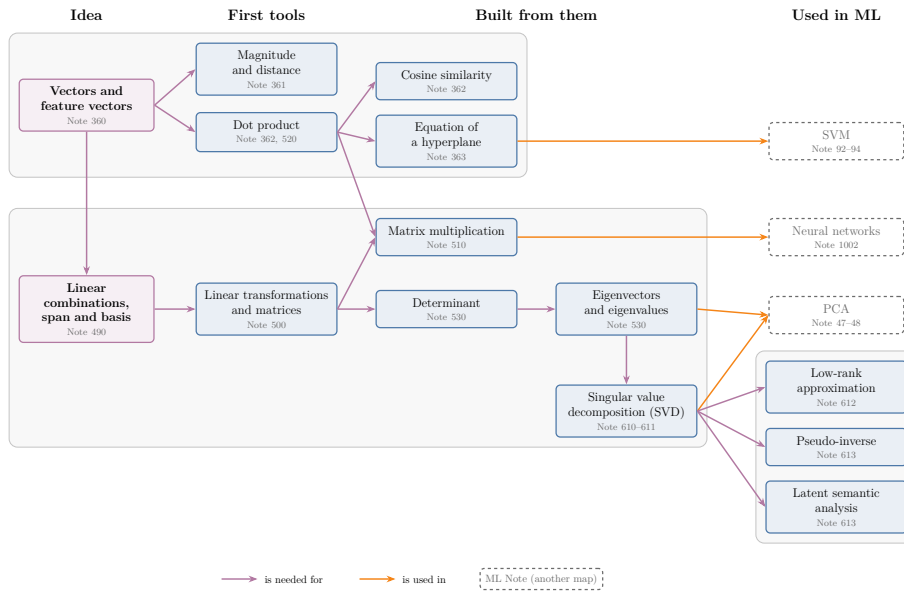


Figure 11: Concept map: Linear algebra

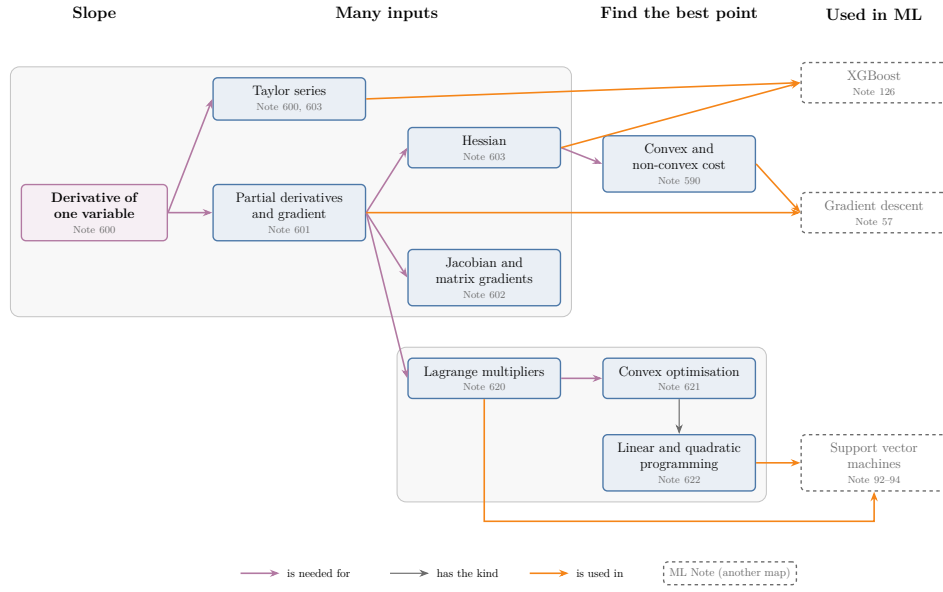


Figure 12: Concept map: Calculus and optimisation

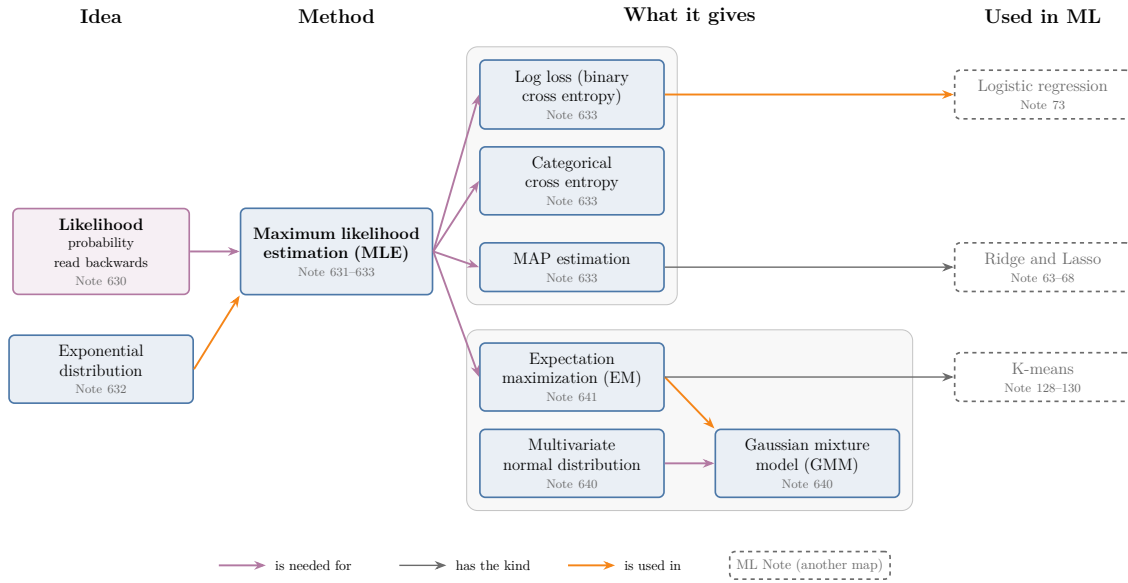


Figure 13: Concept map: Likelihood, MLE and mixture models

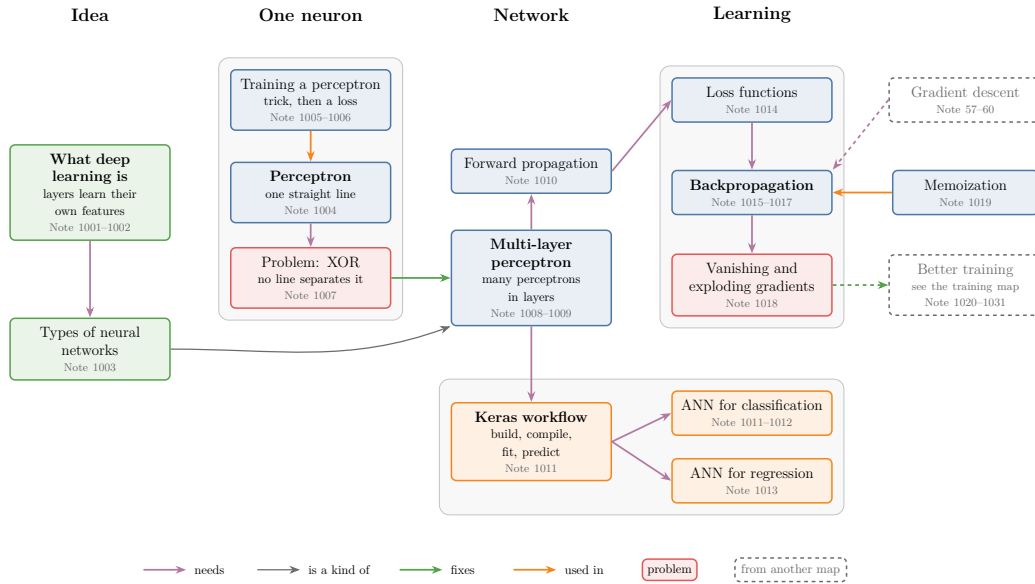


Figure 14: Concept map: Deep learning: perceptron to backpropagation

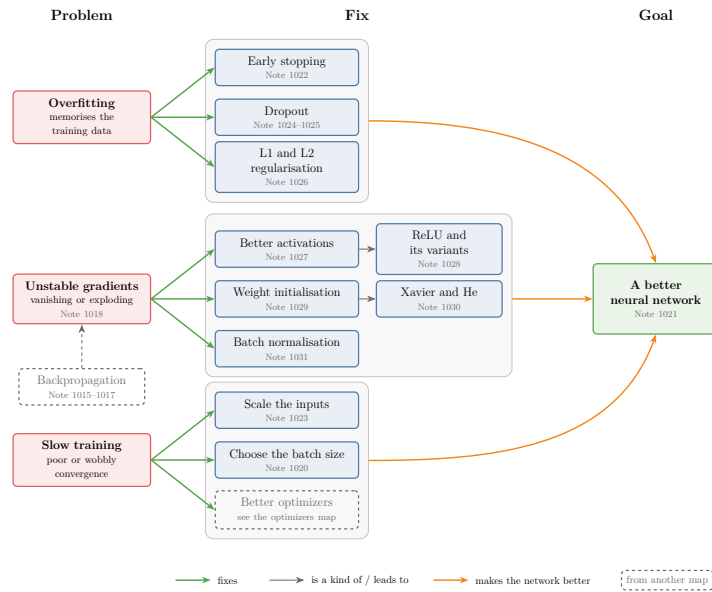


Figure 15: Concept map: Deep learning: training better networks

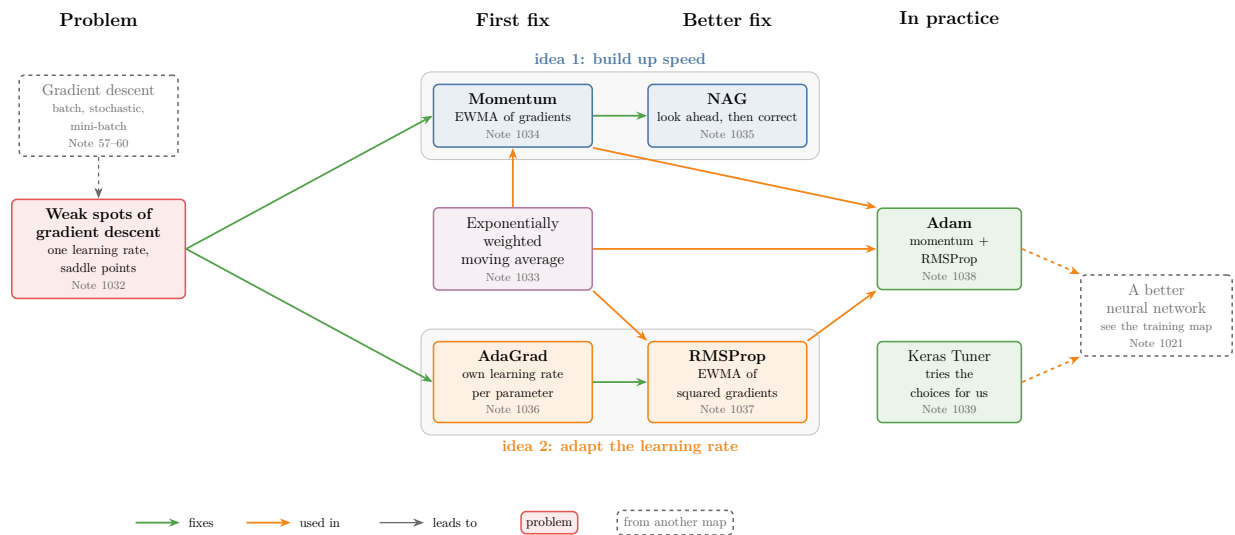


Figure 16: Concept map: Deep learning: optimizers

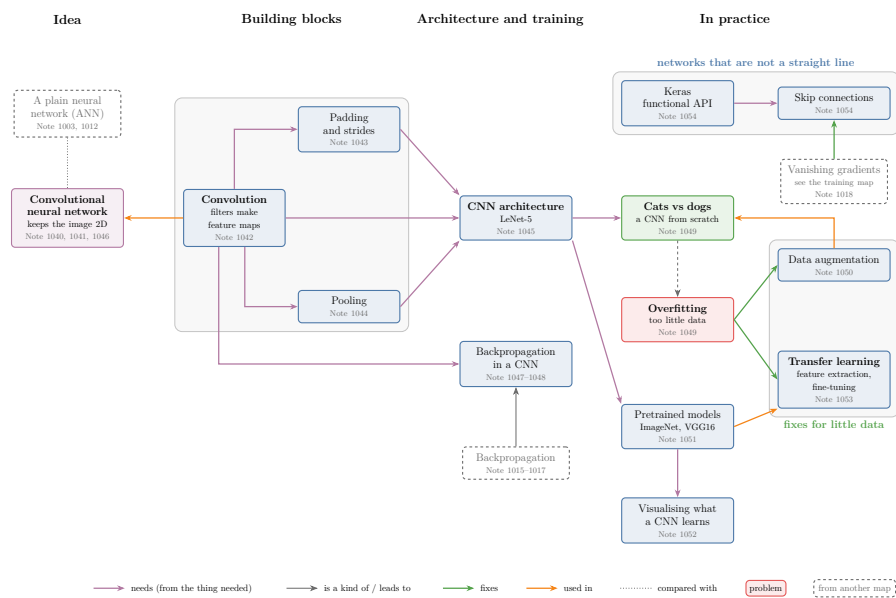


Figure 17: Concept map: Deep learning: convolutional networks

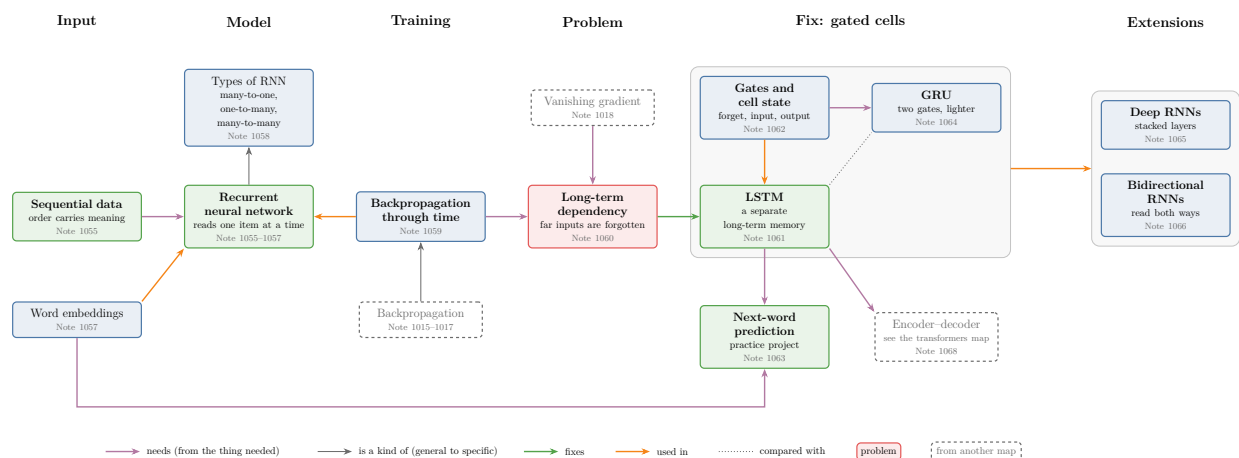


Figure 18: Concept map: Deep learning: recurrent networks

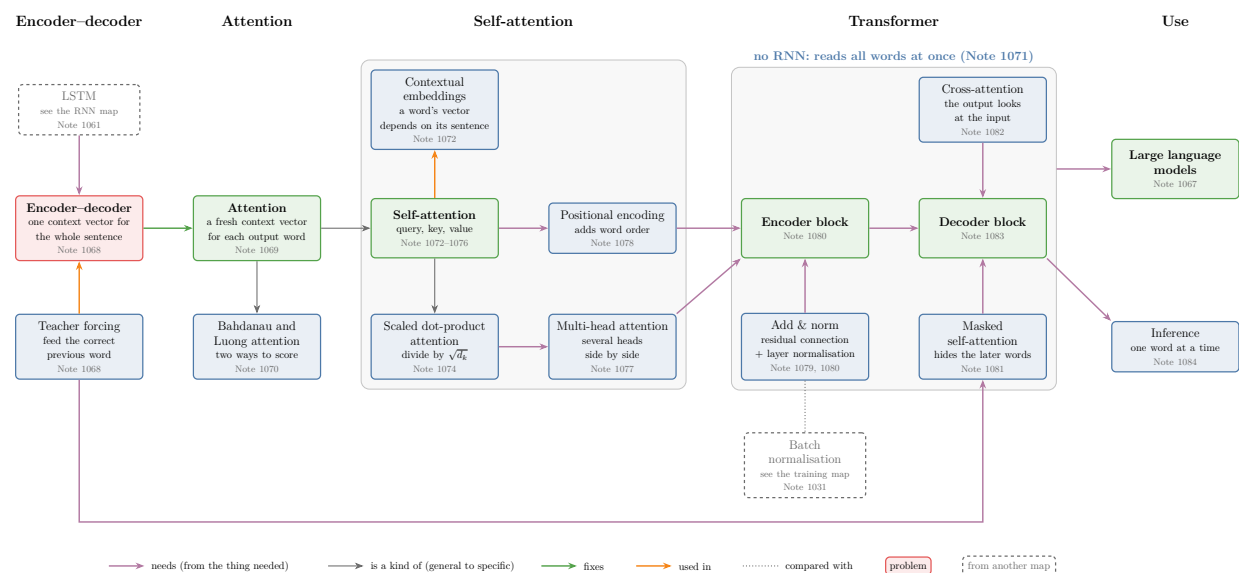


Figure 19: Concept map: Deep learning: LLMs, attention and transformers

4. The Learning path

Key point: Before each Note, read the Notes it builds on.

A Note is easiest to read when the ideas it uses are already familiar. The Notes that teach those ideas are its **prerequisites**: the Notes to read first. Each prerequisite has prerequisites of its own, so the reading order grows backwards from the Note we want, one round at a time.

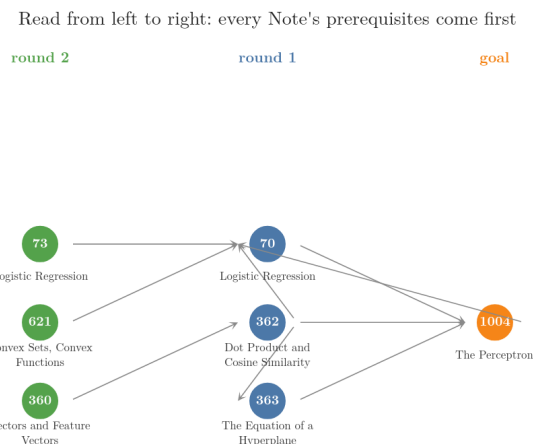


Figure 20: The reading order for the perceptron Note (DL-004), grown backwards. Round 1 adds the three Notes it builds on; round 2 adds the Notes that those build on. Arrows point from a prerequisite to the Note that needs it.

Figure 20 builds the reading order for the perceptron Note (Note DL-004) step by step:

1. **Goal.** We want to read Note DL-004.
2. **Round 1.** Its row in the table below lists Note ML-069, Note MA-050 and Note MA-051.
3. **Round 2.** Each of those has its own row: Note ML-069 builds on Note ML-072, Note MA-067; Note MA-050 builds on Note MA-048; Note MA-051 builds on nothing new.
4. **Reading.** Read the picture from left to right: green Notes first, then blue, then the goal. Every arrow points from a Note to a Note that needs it.

Each row of the table lists a Note's Concepts and the Notes to read first, with at most the four most recent. A row comes from the **needs**, **is a kind of**, **fixes** and **used in** Links of section 3: when a Concept of the Note builds on another Concept, the latest earlier Note that teaches that other Concept is read first. Notes marked *coming* or *deferred* are not written yet.

No.	Concepts	Read first	Note
ML-001	Data mining, Machine learning	Note ML-002, Note ML-010, Note ML-011, Note MA-001	written

No.	Concepts	Read first	Note
ML-002	Artificial intelligence, Deep learning, Features, Machine learning, Neural networks, Symbolic AI and expert systems	Note ML-011 , Note MA-001 , Note MA-053 , Note MA-054	written
ML-003	Anomaly detection, Association rule learning, Classification problems, Clustering, Dimensionality reduction, Feature construction and splitting, Labelled data, PCA, Regression problems, Reinforcement learning, Semi-supervised learning, Supervised learning, Unsupervised learning	Note ML-126 , Note MA-053 , Note MA-056 , Note MA-057	written
ML-004	Batch (offline) learning, Deployment, Model drift, Retraining	Note ML-007 , Note ML-009 , Note ML-012	written
ML-005	Online learning, Out-of-core learning, Retraining, Stochastic gradient descent	Note ML-004 , Note ML-056 , Note ML-059	written
ML-006	Feature scaling, Instance-based learning, K-nearest neighbours, Model-based learning	Note ML-022 , Note ML-028 , Note ML-045 , Note MA-049	written

No.	Concepts	Read first	Note
ML-007	APIs, Deployment, Enough data, Feature construction and splitting, Feature engineering, Labelled data, MLOps and cost, Missing values, Outliers, Overfitting, Poor-quality data, Sampling noise and bias, Software integration, Underfitting, Web scraping	Note ML-015 , Note ML-022 , Note ML-028 , Note ML-040	written
ML-008	Applications of ML, Data mining	Note ML-001 , Note ML-003	written
ML-009	APIs, Beta and A/B testing, Bivariate and multivariate analysis, CSV files, Deployment, Ensemble learning, Exploratory data analysis, Feature construction and splitting, Feature engineering, Feature scaling, Feature selection, Framing an ML problem, Hyperparameter tuning, Imbalanced data, ML development life cycle, MLOps and cost, Missing values, Model drift, Outliers, Poor-quality data, Retraining, Saving models with pickle, Standardization, Univariate analysis, Web scraping	Note ML-093 , Note MA-038 , Note MA-045 , Note MA-046	written

No.	Concepts	Read first	Note
ML-010	Features, One-hot encoding, Tensors	Note ML-022 , Note ML-026	written
ML-011	Setup: conda, Jupyter and Colab	nothing	written
ML-012	Accuracy, CSV files, Data leakage, Deployment, Exploratory data analysis, Feature scaling, Feature selection, Logistic regression, ML development life cycle, ML pipelines, Saving models with pickle, Standardization, Train-test split	Note MA-045 , Note MA-046 , Note MA-067 , Note DL-004	written
ML-013	Framing an ML problem	Note ML-003 , Note ML-004 , Note ML-005	written
ML-014	CSV files	nothing	written
ML-015	JSON and SQL data	nothing	written
ML-016	APIs	Note ML-015	written
ML-017	Web scraping	nothing	written
ML-018	Correlation, Descriptive statistics, Exploratory data analysis, Variance	Note ML-009 , Note ML-021 , Note MA-006 , Note MA-007	written
ML-019	Kernel density estimation (KDE), Outliers, Probability density function (PDF), Skewness, Univariate analysis	Note ML-018 , Note ML-040 , Note MA-003 , Note MA-023	written
ML-020	Bivariate and multivariate analysis, Correlation	Note ML-009 , Note ML-018	written
ML-021	Kurtosis and moments, Pandas Profiling	Note ML-011	written

No.	Concepts	Read first	Note
ML-022	Binning and binarization, Encoding categorical data, Feature construction and splitting, Feature engineering, Feature extraction, Feature scaling, Feature selection, Feature transformation, Missing values, One-hot encoding, Outliers, Simple imputation (mean, median, mode, constant)	Note ML-066 , Note ML-093 , Note MA-045 , Note MA-046	written
ML-023	Feature scaling, Standardization	Note ML-007 , Note ML-012 , Note ML-018 , Note ML-022	written
ML-024	Normalization	Note ML-006	written
ML-025	Encoding categorical data, Ordinal and label encoding	Note ML-012 , Note ML-022	written
ML-026	Multicollinearity, One-hot encoding	Note ML-022	written
ML-027	Column transformer	Note ML-022 , Note ML-029	written
ML-028	Chi-square tests, Cross-validation, Deployment, Grid and random search, ML pipelines, Saving models with pickle	Note ML-022 , Note ML-027 , Note MA-038 , Note MA-014	written
ML-029	Cross-validation, Function transformer, Q-Q plot	Note ML-021 , Note ML-022 , Note MA-020 , Note MA-022	written
ML-030	Power transformer	Note ML-019 , Note ML-022 , Note ML-029	written

No.	Concepts	Read first	Note
ML-031	Binning and binarization, K-means	Note ML-007 , Note ML-022 , Note ML-122 , Note MA-074	written
ML-032	Mixed variables	Note ML-003 , Note ML-007	written
ML-033	Date and time features	Note ML-007 , Note ML-009	written
ML-034	Complete case analysis, Missing values	Note ML-007 , Note ML-009	written
ML-035	Missing values, Simple imputation (mean, median, mode, constant)	Note ML-007 , Note ML-012	written
ML-036	Missing values, Simple imputation (mean, median, mode, constant)	Note ML-007 , Note ML-012	written
ML-037	Grid and random search, ML pipelines, Missing indicator, Missing values, Random sample imputation	Note ML-012 , Note ML-022 , Note ML-027 , Note ML-028	written
ML-038	KNN imputer, Vector magnitude, distance and scalar operations	Note ML-006 , Note ML-007 , Note ML-037 , Note MA-048	written
ML-039	Iterative imputation (MICE)	Note ML-007 , Note ML-022 , Note ML-049	written
ML-040	Capping (winsorization), IQR outlier method, Outliers, Percentile outlier method, Trimming outliers, Z-score outlier method	Note ML-018 , Note ML-019 , Note MA-008 , Note MA-020	written

No.	Concepts	Read first	Note
ML-041	Capping (winsorization), Normal distribution, Trimming outliers, Z-score outlier method	Note ML-018 , Note ML-019 , Note ML-040 , Note MA-003	written
ML-042	Capping (winsorization), IQR outlier method, Trimming outliers	Note ML-007 , Note ML-009 , Note ML-018 , Note ML-040	written
ML-043	Capping (winsorization), Percentile outlier method, Trimming outliers	Note ML-007 , Note MA-008	written
ML-044	Feature construction and splitting	Note ML-007 , Note ML-022 , Note ML-028	written
ML-045	Curse of dimensionality, Dimensionality reduction, Feature extraction, Feature selection	Note ML-066 , Note ML-093 , Note MA-045 , Note MA-046	written
ML-046	Feature extraction, PCA, Variance	Note MA-006 , Note MA-053 , Note MA-056 , Note MA-057	written
ML-047	Covariance and covariance matrix, Dot product, Eigenvectors and eigenvalues, Linear transformations and matrices, PCA	Note MA-048 , Note MA-052 , Note MA-056 , Note MA-057	written
ML-048	PCA	Note ML-047 , Note MA-053 , Note MA-056 , Note MA-057	written
ML-049	Best-fit line and squared error, Simple linear regression	Note ML-050 , Note ML-051 , Note ML-055 , Note ML-056	written

No.	Concepts	Read first	Note
ML-050	Best-fit line and squared error, Derivatives of one variable, Ordinary least squares (closed form), Simple linear regression	Note ML-047 , Note ML-051 , Note ML-055 , Note ML-056	written
ML-051	Regression metrics	Note ML-049	written
ML-052	Equation of a hyperplane, Multiple linear regression	Note ML-049 , Note ML-053 , Note ML-057 , Note MA-050	written
ML-053	Multiple linear regression, Normal equation	Note ML-050 , Note ML-057 , Note MA-052 , Note MA-063	written
ML-054	Multiple linear regression, Normal equation	Note ML-050 , Note ML-057 , Note MA-052 , Note MA-063	written
ML-055	Assumptions of linear regression	Note ML-026 , Note ML-029	written
ML-056	Convex and non-convex loss, Gradient descent, Learning rate, Partial derivatives and gradients	Note ML-006 , Note ML-049 , Note MA-061 , Note MA-064	written
ML-057	Batch gradient descent	Note ML-056	written
ML-058	Stochastic gradient descent	Note ML-056	written
ML-059	Mini-batch gradient descent	Note ML-056	written
ML-060	Overfitting, Polynomial features, Polynomial regression, Underfitting	Note ML-012 , Note ML-052	written
ML-061	Bias-variance trade-off	Note ML-007 , Note ML-060	written

No.	Concepts	Read first	Note
ML-062	Regularisation, Ridge regression	Note ML-061 , Note MA-049 , Note MA-066 , Note MA-072	written
ML-063	Ridge regression	Note ML-061 , Note ML-062 , Note MA-066 , Note MA-072	written
ML-064	Ridge regression	Note ML-061 , Note ML-062 , Note MA-066 , Note MA-072	written
ML-065	Ridge regression	Note ML-061 , Note ML-062 , Note MA-066 , Note MA-072	written
ML-066	Lasso regression	Note ML-062 , Note MA-066 , Note MA-072	written
ML-067	Lasso regression	Note ML-062 , Note MA-066 , Note MA-072	written
ML-068	Elastic Net	Note ML-026 , Note ML-062 , Note ML-066	written
ML-069	Equation of a hyperplane, Logistic regression, Perceptron trick	Note ML-072 , Note MA-050 , Note MA-067 , Note DL-004	written
ML-070	Logistic regression, Perceptron trick	Note ML-072 , Note MA-051 , Note MA-067 , Note DL-004	written
ML-071	Logistic regression, Sigmoid function	Note ML-072 , Note MA-067 , Note DL-004 , Note DL-027	written
ML-072	Log loss (binary cross entropy), Logistic regression	Note MA-067 , Note MA-070 , Note DL-004 , Note DL-014	written
ML-073	Sigmoid function	Note DL-027	written

No.	Concepts	Read first	Note
ML-074	Logistic regression	Note ML-071 , Note ML-072 , Note MA-067 , Note DL-004	written
ML-075	Accuracy, Confusion matrix	Note ML-009 , Note ML-012	written
ML-076	Precision, recall and F1	Note ML-009 , Note ML-075	written
ML-077	ROC curve and AUC	Note ML-075 , Note ML-076	written
ML-078	Softmax regression	Note ML-010 , Note ML-012	written
ML-079	Polynomial features	nothing	written
ML-080	Hyperparameter tuning	Note ML-007	written
MA-015	Conditional probability	Note MA-010 , Note MA-014	written
MA-016	Independent and mutually exclusive events	Note MA-015	written
MA-017	Independent and mutually exclusive events	Note MA-015	written
MA-018	Bayes' theorem	Note MA-015 , Note MA-014	written
MA-019	Bayes' theorem	Note MA-015 , Note MA-014	written
ML-081	Naive Bayes	Note ML-019 , Note MA-016 , Note MA-018 , Note MA-020	written
ML-082	Naive Bayes	Note ML-019 , Note MA-016 , Note MA-018 , Note MA-020	written
ML-083	Naive Bayes	Note ML-019 , Note MA-016 , Note MA-018 , Note MA-020	written
ML-084	Naive Bayes, Normal distribution, Probability density function (PDF)	Note ML-003 , Note MA-016 , Note MA-018 , Note MA-003	written

No.	Concepts	Read first	Note
ML-085	Accuracy, Cross-validation, Curse of dimensionality, Data leakage, Decision surface and boundary, Grid and random search, Hyperparameter tuning, K-nearest neighbours, Overfitting, Underfitting	Note ML-006 , Note ML-012 , Note ML-022 , Note MA-049	written
ML-086	Support vector machines	Note MA-050 , Note MA-051 , Note MA-066 , Note MA-068	written
ML-087	Support vector machines	Note MA-050 , Note MA-051 , Note MA-066 , Note MA-068	written
ML-088	Hinge loss and soft margin, Support vector machines	Note MA-050 , Note MA-051 , Note MA-066 , Note MA-068	written
ML-089	Kernel trick	nothing	written
ML-090	Kernel trick	nothing	written
ML-091	Decision trees, Entropy, information gain and Gini	Note ML-006 , Note ML-085	written
ML-092	Decision trees, Hyperparameter tuning	Note ML-006 , Note ML-007 , Note ML-085 , Note ML-091	written
ML-093	Feature importance, Grid and random search, Regression trees	Note ML-022 , Note ML-028 , Note ML-051 , Note ML-091	written
ML-094	Decision trees, Feature importance	Note ML-006 , Note ML-085 , Note ML-091	written

No.	Concepts	Read first	Note
ML-095	Boosting, Ensemble learning	Note ML-007 , Note ML-061 , Note ML-091	written
ML-096	Bernoulli and binomial distributions, Voting ensembles	Note ML-009 , Note ML-028 , Note MA-016 , Note MA-003	written
ML-097	Cross-validation, Voting ensembles	Note ML-009 , Note MA-016 , Note MA-021	written
ML-098	Cross-validation, Voting ensembles	Note ML-009 , Note MA-016 , Note MA-021	written
ML-099	Bagging, OOB score	Note ML-028 , Note ML-061 , Note ML-091 , Note MA-012	written
ML-100	Bagging, Grid and random search, OOB score	Note ML-028 , Note ML-061 , Note ML-091 , Note MA-012	written
ML-101	Bagging, Grid and random search, OOB score	Note ML-028 , Note ML-061 , Note ML-091 , Note MA-012	written
ML-102	Random forest	Note ML-028 , Note ML-091 , Note ML-093 , Note ML-099	written
ML-103	Bias-variance trade-off, Random forest	Note ML-060 , Note ML-091 , Note ML-093 , Note ML-099	written
ML-104	Random forest	Note ML-028 , Note ML-091 , Note ML-093 , Note ML-099	written
ML-105	Hyperparameter tuning, Random forest	Note ML-028 , Note ML-091 , Note ML-093 , Note ML-099	written

No.	Concepts	Read first	Note
ML-106	Cross-validation, Grid and random search, Random forest	Note ML-022 , Note ML-091 , Note ML-093 , Note ML-099	written
ML-107	OOB score, Random forest	Note ML-028 , Note ML-091 , Note ML-093 , Note ML-099	written
ML-108	Feature importance, Permutation importance, Random forest	Note ML-009 , Note ML-028 , Note ML-091 , Note ML-099	written
ML-109	AdaBoost	Note ML-009 , Note ML-091 , Note ML-095	written
ML-110	AdaBoost	Note ML-009 , Note ML-091 , Note ML-095	written
ML-111	AdaBoost	Note ML-009 , Note ML-091 , Note ML-095	written
ML-112	AdaBoost, Cross-validation, Grid and random search, Hyperparameter tuning, Learning rate	Note ML-012 , Note ML-022 , Note ML-091 , Note ML-095	written
ML-113	Boosting	Note ML-007 , Note ML-009 , Note ML-061	written
ML-114	Gradient boosting, Learning rate	Note ML-071 , Note ML-072 , Note ML-093 , Note ML-095	written
ML-115	Gradient boosting	Note ML-071 , Note ML-072 , Note ML-093 , Note ML-095	written
ML-116	Gradient boosting	Note ML-071 , Note ML-072 , Note ML-093 , Note ML-095	written

No.	Concepts	Read first	Note
ML-117	Missing values, XGBoost	Note ML-062 , Note ML-114 , Note MA-061 , Note MA-064	written
ML-118	XGBoost	Note ML-062 , Note ML-114 , Note MA-061 , Note MA-064	written
ML-119	XGBoost	Note ML-062 , Note ML-114 , Note MA-061 , Note MA-064	written
ML-120	Hessian and multivariate Taylor, Taylor series, XGBoost	Note ML-114 , Note MA-056 , Note MA-061 , Note MA-062	written
ML-121	Cross-validation, Stacking and blending	Note ML-009	written
ML-122	Clustering, Elbow method and WCSS, K-means	Note ML-003 , Note ML-006 , Note MA-074	written
ML-123	Clustering, Elbow method and WCSS, K-means	Note ML-003 , Note ML-006 , Note MA-074	written
ML-124	Clustering, K-means	Note ML-003 , Note ML-006 , Note ML-122 , Note MA-074	written
ML-125	Clustering, Hierarchical clustering	Note ML-003	written
ML-126	Anomaly detection, Clustering, DBSCAN	Note ML-003	written
ML-127	Balanced random forest, Cost-sensitive learning, Imbalanced data, Random under- and oversampling, SMOTE	Note ML-028 , Note ML-072 , Note ML-077 , Note ML-102	written
ML-128	Bayesian optimisation, Optuna	Note ML-009 , Note ML-028	written

No.	Concepts	Read first	Note
MA-003	Descriptive statistics, Inferential statistics, Probability distributions	Note MA-004 , Note MA-007 , Note MA-020 , Note MA-035	written
MA-004	Discrete and continuous data, Inferential statistics, Population, sample, parameter and statistic	Note ML-007 , Note MA-035	written
MA-005	Descriptive statistics, Measures of central tendency	Note ML-009 , Note MA-007	written
MA-006	Bessel's correction, Descriptive statistics, Variance	Note ML-009 , Note MA-004 , Note MA-007	written
MA-007	Descriptive statistics, Frequency tables	Note ML-009	written
MA-008	Descriptive statistics, Percentiles, quartiles and box plots	Note ML-009 , Note MA-007	written
MA-009	Correlation, Correlation and causation, Covariance and covariance matrix	Note ML-018	written
MA-020	Normal distribution, Probability distributions, Random variables	Note ML-019 , Note MA-004	written
MA-021	Bernoulli and binomial distributions, Cumulative distribution function (CDF), Probability mass function (PMF), Uniform distribution	Note ML-019 , Note MA-003	written

No.	Concepts	Read first	Note
MA-022	Cumulative distribution function (CDF), Log-normal distribution, Poisson distribution, Probability density function (PDF)	Note MA-003 , Note MA-021 , Note MA-012	written
MA-023	Density estimation, Kernel density estimation (KDE)	Note ML-019 , Note MA-004	written
MA-024	Normal distribution	Note ML-019 , Note MA-003	written
MA-025	Standard normal and the z-table, Z-score outlier method	Note ML-018 , Note ML-019 , Note MA-020 , Note MA-021	written
MA-026	Skewness	Note ML-018	written
MA-027	Cumulative distribution function (CDF), Kernel density estimation (KDE)	Note ML-019 , Note MA-021 , Note MA-023	written
MA-028	Kurtosis and moments, Q-Q plot	Note MA-020	written
MA-029	Log-normal distribution, Uniform distribution	Note MA-003	written
MA-030	Pareto distribution and power laws	nothing	written
MA-031	Bernoulli and binomial distributions	Note MA-003	written
MA-033	Central limit theorem, Sampling distribution and standard error	Note MA-004 , Note MA-020	written
MA-034	Central limit theorem	Note MA-020 , Note MA-033	written
MA-035	Confidence intervals	Note MA-025 , Note MA-033 , Note MA-037	written

No.	Concepts	Read first	Note
MA-036	Confidence intervals	Note MA-025 , Note MA-033 , Note MA-037	written
MA-037	Confidence intervals, Student's t-distribution	Note MA-006 , Note MA-025 , Note MA-033	written
MA-038	Hypothesis testing: null and alternative	Note MA-003 , Note MA-033 , Note MA-040 , Note MA-041	written
MA-039	Hypothesis testing: null and alternative, Z-test and rejection regions	Note MA-025 , Note MA-033 , Note MA-040 , Note MA-041	written
MA-040	Beta and A/B testing, Type I and II errors, power, tails	Note ML-004 , Note MA-038	written
MA-041	P-values	Note MA-021	written
MA-042	T-tests: one-sample, two-sample, paired	Note MA-037 , Note MA-038	written
MA-043	T-tests: one-sample, two-sample, paired	Note MA-037 , Note MA-038	written
MA-010	Events and sample spaces	nothing	written
MA-011	Empirical vs theoretical probability, probability rules	Note MA-010	written
MA-012	Expected value and variance of a random variable	Note MA-020	written
MA-013	Empirical vs theoretical probability, probability rules, Venn diagrams and contingency tables	Note MA-010	written
MA-014	Bayes' theorem, Conditional probability, Joint and marginal probability	Note MA-010 , Note MA-013	written
MA-047	Linear algebra roadmap	nothing	written

No.	Concepts	Read first	Note
MA-048	Bag of words, Vectors and feature vectors	Note ML-010	written
MA-049	Vector magnitude, distance and scalar operations	Note MA-048	written
MA-050	Cosine similarity, Dot product	Note MA-048	written
MA-051	Equation of a hyperplane	Note MA-050	written
MA-001	Role of mathematics in ML	nothing	written
MA-052	Linear combinations, span and basis	Note MA-048	written
MA-053	Linear transformations and matrices	Note MA-052	written
MA-054	Matrix multiplication as composition	Note MA-050 , Note MA-053	written
MA-055	Dot product	Note MA-048	written
MA-056	Determinant, Eigenvectors and eigenvalues	Note MA-053	written
MA-032	Poisson distribution	Note MA-021 , Note MA-012	written
MA-044	Choosing a hypothesis test, Correlation significance test, One-sample proportion test, T-tests: one-sample, two-sample, paired	Note ML-018 , Note MA-037 , Note MA-038 , Note MA-039	written
MA-045	Chi-square tests	Note MA-038 , Note MA-014	written
MA-046	One-way ANOVA	Note ML-018 , Note MA-038	written
MA-002	How to learn the maths for ML	Note MA-001	written
MA-065	Convex and non-convex loss	Note MA-064	written

No.	Concepts	Read first	Note
MA-061	Derivatives of one variable, Taylor series	nothing	written
MA-062	Partial derivatives and gradients	Note MA-061	written
MA-063	Jacobian and matrix gradients	Note MA-053 , Note MA-054 , Note MA-062	written
MA-064	Hessian and multivariate Taylor, Taylor series	Note MA-056 , Note MA-061 , Note MA-062	written
MA-057	Singular value decomposition	Note MA-053 , Note MA-056	written
MA-058	Singular value decomposition	Note MA-053 , Note MA-056	written
MA-059	Low-rank approximation (truncated SVD)	Note MA-057	written
MA-060	Latent semantic analysis, Moore-Penrose pseudo-inverse	Note ML-026 , Note MA-048 , Note MA-057	written
MA-066	Lagrange multipliers, KKT and duality	Note MA-062	written
MA-067	Convex sets and convex optimisation	Note ML-056 , Note MA-066	written
MA-068	Linear and quadratic programming	Note MA-067	written
MA-069	Likelihood, Normal distribution	Note ML-019 , Note MA-003	written
MA-070	Maximum likelihood estimation (MLE)	Note MA-061 , Note MA-062 , Note MA-069 , Note MA-071	written
MA-071	Exponential distribution, Maximum likelihood estimation (MLE)	Note ML-019 , Note MA-061 , Note MA-062 , Note MA-069	written

No.	Concepts	Read first	Note
MA-072	Categorical and sparse categorical cross-entropy, Log loss (binary cross entropy), MAP estimation, Maximum likelihood estimation (MLE)	Note MA-062 , Note MA-069 , Note MA-071 , Note DL-014	written
MA-073	Gaussian mixture model (GMM), Multivariate normal distribution	Note MA-018 , Note MA-020 , Note MA-070 , Note MA-074	written
MA-074	Expectation maximization (EM), Gaussian mixture model (GMM), K-means	Note MA-018 , Note ML-122 , Note MA-070 , Note MA-073	written
DL-001	What deep learning is	Note ML-001	written
DL-002	Neural networks, Representation learning, What deep learning is	Note ML-001 , Note MA-053 , Note MA-054	written
DL-003	History of deep learning, Multi-layer perceptron (MLP), Types of neural networks, Universal approximation theorem	Note DL-007 , Note DL-008 , Note DL-015 , Note DL-027	written
DL-004	Perceptron	Note ML-069 , Note MA-050 , Note MA-051	written
DL-005	Perceptron trick	Note MA-051	written
DL-006	Perceptron loss	Note ML-005 , Note ML-069	written
DL-007	Problem with the perceptron (XOR)	Note DL-004	written
DL-008	MLP notation and parameter count	nothing	written

No.	Concepts	Read first	Note
DL-009	Multi-layer perceptron (MLP), Universal approximation theorem	Note DL-007 , Note DL-008 , Note DL-015 , Note DL-027	written
DL-010	Forward propagation	Note MA-054 , Note DL-008	written
DL-011	ANN for classification, Keras workflow, Training curves (History)	Note ML-078 , Note MA-072 , Note DL-003 , Note DL-015	written
DL-012	ANN for classification, Categorical and sparse categorical cross-entropy, Keras workflow, Training curves (History)	Note MA-070 , Note DL-003 , Note DL-014 , Note DL-015	written
DL-013	ANN for regression, Keras workflow, Training curves (History)	Note ML-024 , Note ML-051 , Note DL-003 , Note DL-015	written
DL-014	Categorical and sparse categorical cross-entropy, Huber loss, Log loss (binary cross entropy), Loss functions in deep learning	Note ML-010 , Note ML-078 , Note MA-070 , Note DL-010	written
DL-015	Backpropagation	Note DL-010 , Note DL-014 , Note DL-019 , Note DL-020	written
DL-016	Backpropagation	Note DL-010 , Note DL-014 , Note DL-019 , Note DL-020	written
DL-017	Backpropagation, Convex and non-convex loss, Gradient descent, Learning rate	Note DL-010 , Note DL-014 , Note DL-019 , Note DL-020	written

No.	Concepts	Read first	Note
DL-018	Exploding gradient and gradient clipping, ReLU, Sigmoid function, Vanishing gradient	Note DL-011 , Note DL-015 , Note DL-027	written
DL-019	Backpropagation, Memoization	Note MA-063 , Note DL-010 , Note DL-014 , Note DL-020	written
DL-020	Batch gradient descent, Batch size in Keras, Gradient descent, Mini-batch gradient descent, Stochastic gradient descent	Note ML-049 , Note ML-056 , Note MA-061 , Note MA-062	written
DL-021	Early stopping, Hyperparameter tuning, Improving a neural network	Note DL-029 , Note DL-031 , Note DL-032 , Note DL-039	written
DL-022	Early stopping, Keras workflow, Training curves (History)	Note ML-007 , Note DL-003	written
DL-023	Feature scaling, Scaling inputs for neural networks, Standardization	Note ML-018 , Note ML-022 , Note ML-024 , Note ML-056	written
DL-024	Dropout	Note ML-007 , Note ML-062	written
DL-025	Dropout, Keras workflow	Note ML-007 , Note ML-062 , Note DL-003	written
DL-026	Keras workflow, L1 and L2 regularisation in neural networks, Overfitting, Regularisation	Note ML-012 , Note ML-056 , Note MA-049 , Note DL-003	written
DL-027	Activation functions, ReLU, Sigmoid function, Tanh	Note DL-004 , Note DL-018	written

No.	Concepts	Read first	Note
DL-028	Dying ReLU problem, Leaky ReLU, PReLU, ELU and SELU, ReLU	Note DL-018 , Note DL-027	written
DL-029	Exploding gradient and gradient clipping, Vanishing gradient, Weight initialisation	Note ML-071 , Note DL-011 , Note DL-015	written
DL-030	Vanishing gradient, Weight initialisation, Xavier and He initialisation	Note DL-011 , Note DL-015 , Note DL-018 , Note DL-027	written
DL-031	Batch normalisation, Covariate shift, Standardization	Note ML-012 , Note ML-018 , Note ML-059 , Note DL-033	written
DL-032	Local minima and saddle points, Optimizers in deep learning	Note ML-056 , Note ML-059 , Note MA-067 , Note DL-015	written
DL-033	Exponentially weighted moving average (EWMA)	nothing	written
DL-034	Local minima and saddle points, SGD with momentum	Note MA-067 , Note DL-032 , Note DL-033	written
DL-035	Nesterov accelerated gradient (NAG)	Note DL-032 , Note DL-034	written
DL-036	AdaGrad	Note ML-056 , Note DL-032	written
DL-037	RMSProp	Note DL-032 , Note DL-033 , Note DL-036	written
DL-038	Adam	Note DL-032 , Note DL-033 , Note DL-034 , Note DL-037	written
DL-039	Hyperparameter tuning, Keras Tuner	Note ML-007 , Note DL-011	written
DL-040	Convolutional neural network (CNN)	Note DL-002 , Note DL-042 , Note DL-044	written
DL-041	Convolutional neural network (CNN)	Note DL-002 , Note DL-042 , Note DL-044	written

No.	Concepts	Read first	Note
DL-042	Convolution operation and feature maps	nothing	written
DL-043	Padding and strides	Note DL-042	written
DL-044	Pooling	Note DL-042	written
DL-045	CNN architecture (LeNet-5)	Note DL-042 , Note DL-043 , Note DL-044	written
DL-046	Convolutional neural network (CNN)	Note DL-002 , Note DL-042 , Note DL-044	written
DL-047	Backpropagation in a CNN	Note DL-015 , Note DL-042	written
DL-048	Backpropagation in a CNN	Note DL-015 , Note DL-042	written
DL-049	Image classification with a CNN (cats vs dogs)	Note DL-011 , Note DL-023 , Note DL-045 , Note DL-050	written
DL-050	Data augmentation	Note ML-007 , Note ML-062	written
DL-051	Pretrained models and ImageNet	Note ML-003 , Note DL-045	written
DL-052	Visualising what a CNN learns	Note DL-042 , Note DL-051	written
DL-053	Transfer learning (feature extraction and fine-tuning)	Note ML-007 , Note DL-051	written
DL-054	Keras functional API, Skip connections	Note DL-011 , Note DL-018	written
DL-055	Recurrent neural network (RNN), Sequence padding, Sequential data	Note DL-027 , Note DL-056 , Note DL-057 , Note DL-059	written
DL-056	Parameter sharing across time steps, Recurrent neural network (RNN)	Note DL-027 , Note DL-055 , Note DL-057 , Note DL-059	written
DL-057	Recurrent neural network (RNN), Sequence padding, Tokenization and integer encoding of text, Word embeddings	Note DL-027 , Note DL-055 , Note DL-056 , Note DL-059	written

No.	Concepts	Read first	Note
DL-058	Sequence-to-sequence (encoder-decoder), Types of RNN (many-to-one, one-to-many, many-to-many)	Note DL-055 , Note DL-061 , Note DL-068	written
DL-059	Backpropagation through time (BPTT), Parameter sharing across time steps	Note DL-015	written
DL-060	Exploding gradient and gradient clipping, Long-term dependency problem, Vanishing gradient	Note ML-071 , Note DL-011 , Note DL-015 , Note DL-059	written
DL-061	LSTM (long short-term memory)	Note DL-055 , Note DL-060 , Note DL-062	written
DL-062	LSTM (long short-term memory), LSTM gates (forget, input, output) and cell state	Note DL-055 , Note DL-060	written
DL-063	LSTM (long short-term memory), Next-word prediction with an LSTM	Note DL-055 , Note DL-057 , Note DL-060 , Note DL-062	written
DL-064	GRU (gated recurrent unit)	Note DL-055 , Note DL-062	written
DL-065	Deep (stacked) RNNs	Note DL-055 , Note DL-061 , Note DL-064	written
DL-066	Bidirectional RNNs	Note DL-055 , Note DL-061 , Note DL-064	written
DL-067	Large language models (LLMs)	Note DL-071 , Note DL-087	written
DL-068	Sequence-to-sequence (encoder-decoder), Teacher forcing	Note DL-058 , Note DL-061	written

No.	Concepts	Read first	Note
DL-069	Attention mechanism, Bahdanau (additive) attention	Note DL-058	written
DL-070	Bahdanau (additive) attention, Luong (multiplicative) attention	Note DL-069	written
DL-071	Transformer	Note DL-058 , Note DL-081 , Note DL-084 , Note DL-086	written
DL-073	Contextual embeddings, Self-attention (query, key, value)	Note DL-057 , Note DL-069 , Note DL-072	written
DL-074	Self-attention (query, key, value)	Note DL-069	written
DL-075	Scaled dot-product attention	Note DL-073	written
DL-076	Self-attention (query, key, value)	Note DL-069	written
DL-077	Self-attention (query, key, value)	Note DL-069	written
DL-078	Multi-head attention	Note DL-075	written
DL-079	Positional encoding	Note DL-073	written
DL-080	Layer normalisation	nothing	written
DL-081	Residual connections and add & norm, Transformer, Transformer encoder	Note DL-080 , Note DL-084 , Note DL-086 , Note DL-089	written
DL-082	Masked self-attention, Teacher forcing	Note DL-073	written
DL-083	Cross-attention	nothing	written
DL-084	Transformer, Transformer decoder	Note DL-081 , Note DL-082 , Note DL-083 , Note DL-086	written
DL-085	Transformer inference (autoregressive decoding, KV cache, beam search)	Note DL-084	written

No.	Concepts	Read first	Note
DL-086	Label smoothing, Learning-rate warm-up schedule, The transformer end to end (capstone)	Note DL-038 , Note DL-081 , Note DL-084 , Note DL-085	written
DL-072	Meaning as direction in embedding space	Note MA-050 , Note DL-057	written
DL-087	Decoder-only GPT, GELU activation	Note DL-071 , Note DL-082 , Note DL-084 , Note DL-088	written
DL-088	Logit lens, Unembedding, logits, temperature and sampling	Note ML-078	written
DL-089	MLP blocks as fact storage	Note DL-087	written
DL-090	Superposition and nearly perpendicular directions	Note DL-072	written

5. The Algorithm chooser

Key point: Two questions narrow the choice: do we have labelled outputs, and are they numbers or categories?

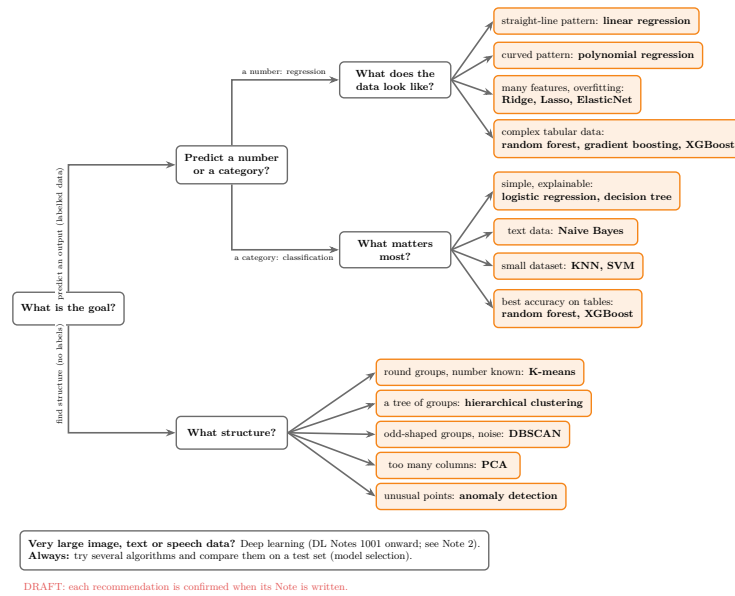


Figure 21: The Algorithm chooser (draft)

Figure 21 is a starting point, not a rule: in practice we try several suitable algorithms and compare them on a test set. It is a draft until the algorithm Notes are written.

6. Sources

Built from - CampusX, “100 Days of Machine Learning”, YouTube playlist (134 videos), https://www.youtube.com/playlist?list=PLKnIA16_Rmvbr7zKYQuBfsVkjoLcJgxHH - CampusX, “100 Days of Deep Learning”, YouTube playlist (84 videos), <https://www.youtube.com/playlist?list=P...> - CampusX, “Maths for Machine Learning”, YouTube playlist (23 sessions), <https://www.youtube.com/playlist?list=PL3vADST...> - Every Note listed in the Learning path: each Note’s own Sources name the video, book or paper it was built from.

Other references - The Concept map data: `course_map/concepts.yaml` in this repository.

7. Key terms

Term	Meaning
Course map (G-2160)	The overview that ties all Notes together, in four views
Pipeline map (G-2161)	Where each Concept sits among the 14 steps of an ML project
Concept map (G-2162)	How Concepts are connected by Links
Learning path (G-2163)	Which Notes to read before which
Algorithm chooser (G-2164)	A flowchart from a problem’s properties to suitable algorithms
Concept (G-2165)	One idea on the Course map
Link (G-2166)	A labelled connection between two Concepts
Draft / confirmed (G-2167)	Guessed from titles / checked against a written Note